

Computer-Aided Geometric Design

—

Tensor-Product Bézier and NURBS Surfaces

Laurent Busé

Centre Inria d'Université Côte d'Azur,
Email: laurent.buse@inria.fr



Bilinear Interpolation

As linear interpolation fits the “simplest” curve between two points, **bilinear interpolation** fits the “simplest” surface between four points.

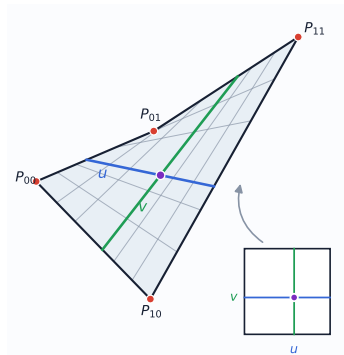
Bilinear Bézier patch

$$\mathbf{P}^{1,1}(u, v) = \sum_{i=0}^1 \sum_{j=0}^1 \mathbf{P}_{ij} B_i^1(u) B_j^1(v).$$

$$(u, v) \in [0, 1]^2$$

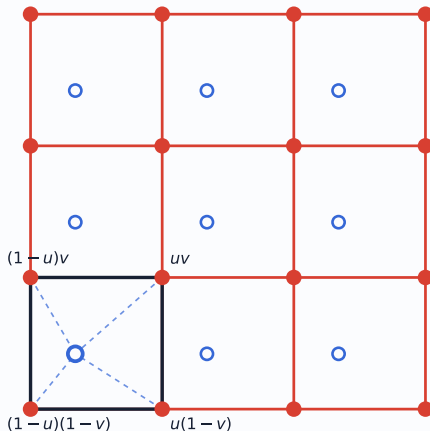
Matrix formulation:

$$\mathbf{P}^{1,1}(u, v) = \begin{bmatrix} 1-u & u \end{bmatrix} \begin{bmatrix} \mathbf{P}_{00} & \mathbf{P}_{01} \\ \mathbf{P}_{10} & \mathbf{P}_{11} \end{bmatrix} \begin{bmatrix} 1-v \\ v \end{bmatrix}.$$



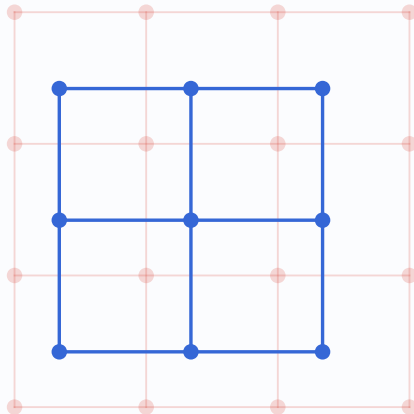
Repeated bilinear interpolation

9 bilinear interpolations — one in each cell



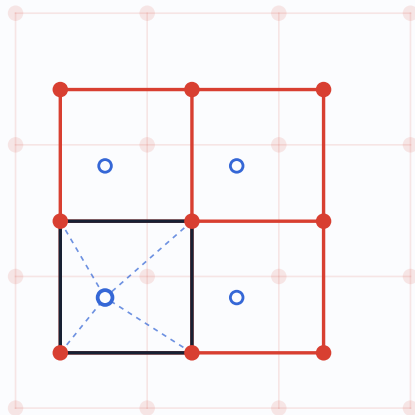
Repeated bilinear interpolation

The 9 results form a new 3×3 grid



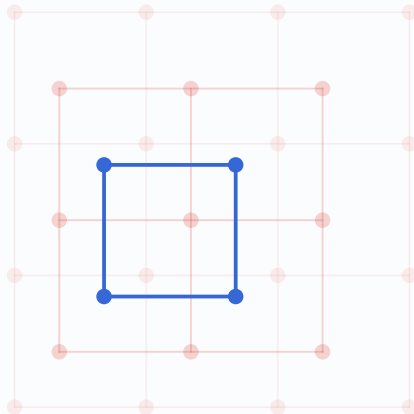
Repeated bilinear interpolation

4 new bilinear interpolations



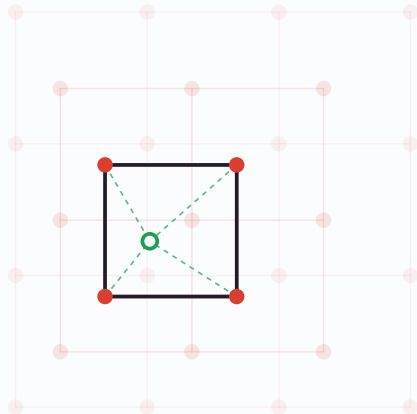
Repeated bilinear interpolation

The 4 results form a new 2×2 grid



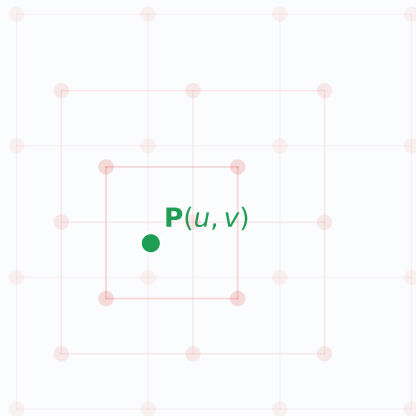
Repeated bilinear interpolation

One last bilinear interpolation



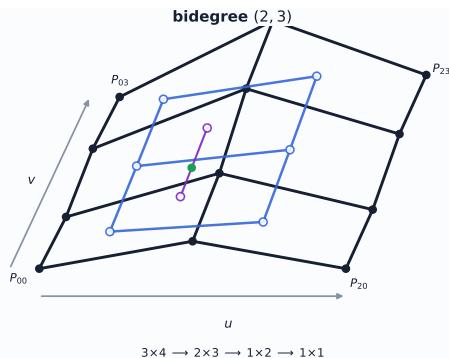
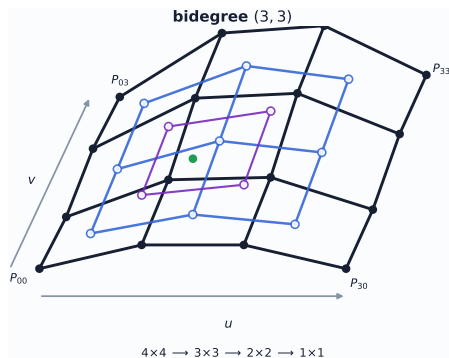
Repeated bilinear interpolation

The final result is a point on the surface



Direct de Casteljau Algorithm

Tensor-product surfaces are obtained from **repeated bilinear interpolation**.



The Tensor-Product Approach

A surface is the locus of a curve that is moving through space.

- Initial Bézier curve of degree m :

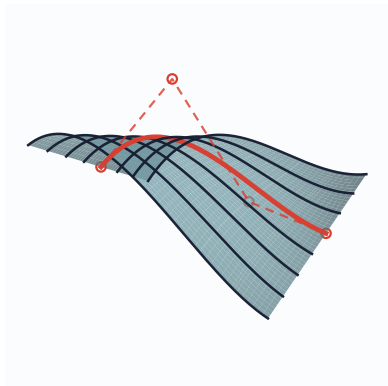
$$\mathbf{P}^m(u) = \sum_{i=0}^m \mathbf{P}_i B_i^m(u).$$

- Each control point moves along a Bézier curve of degree n :

$$\mathbf{P}_i(v) = \sum_{j=0}^n \mathbf{P}_{ij} B_j^n(v).$$

Tensor-product patch

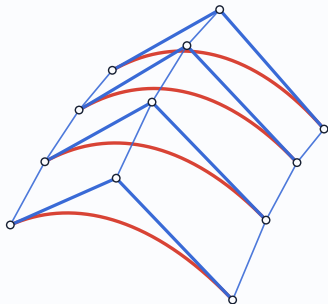
$$\mathbf{P}^{m,n}(u, v) = \sum_{i=0}^m \sum_{j=0}^n \mathbf{P}_{ij} B_i^m(u) B_j^n(v).$$



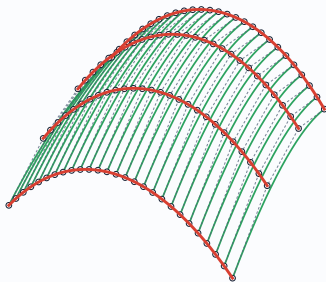
Control Curves and Isoparametric Curves

- ▶ Four **boundary Bézier curves**.
- ▶ The **control curves** are the Bézier curves given by a row or a column of control points.
- ▶ **Isoparametric curves** are Bézier curves on the surface obtained by fixing one parameter.

control curves — bidegree (2, 3)



isoparametric curves



The de Casteljau Algorithm

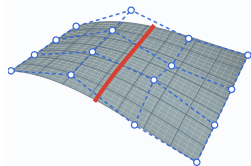
To evaluate a point on a tensor-product Bézier patch corresponding to (s, t) :

- ▶ evaluate all v -control curves at $v = t$;
- ▶ evaluate the resulting isoparametric curve at $u = s$.
- ▶ Alternatively, one can process the u -control curves first.

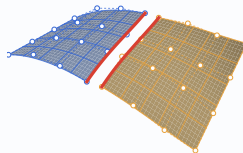
Subdivision

The same algorithm cuts a tensor-product Bézier patch into two pieces, either in the u - or in the v -parameter direction.

one patch, cut along $u = s$

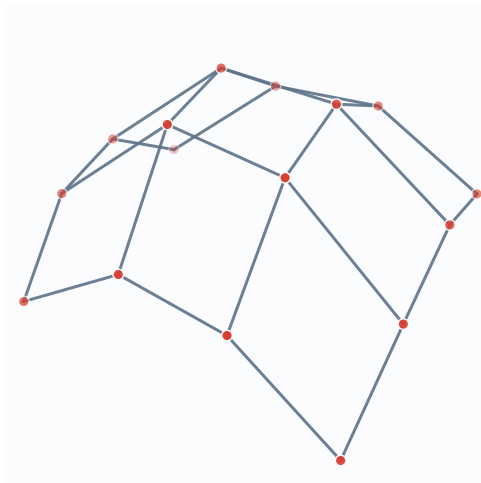


two tensor-product Bézier patches



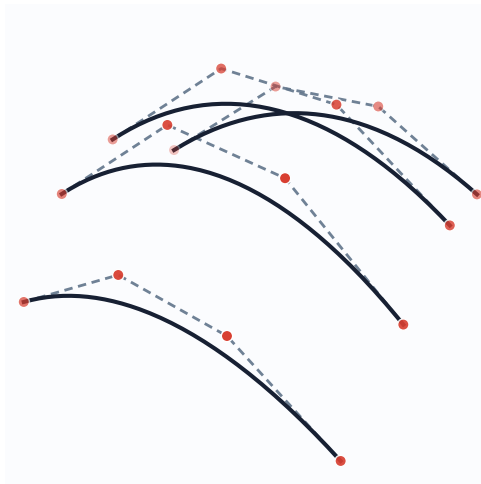
The de Casteljau Algorithm

Start from the bicubic 4×4 control net.



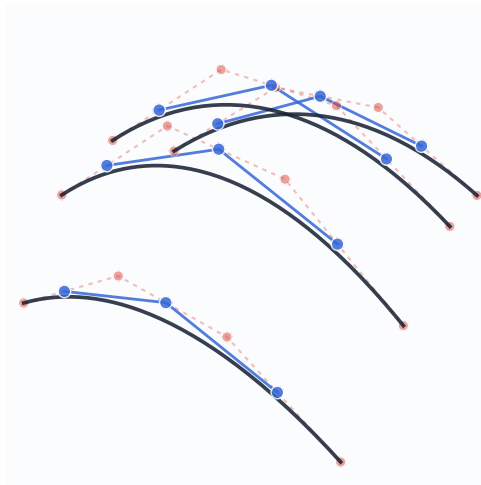
The de Casteljau Algorithm

First order: expose the four control curves in the u -direction.



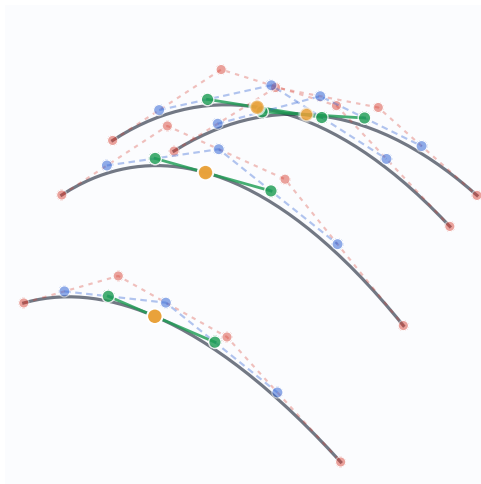
The de Casteljau Algorithm

Apply the first de Casteljau reduction at $u = u_0$ to all four curves.



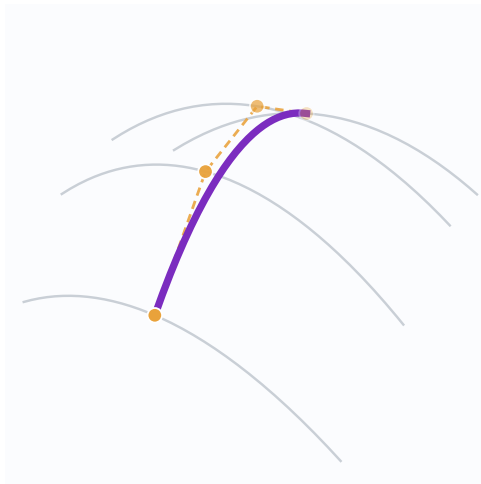
The de Casteljau Algorithm

Complete the four evaluations; the gold points form a transverse control polygon.



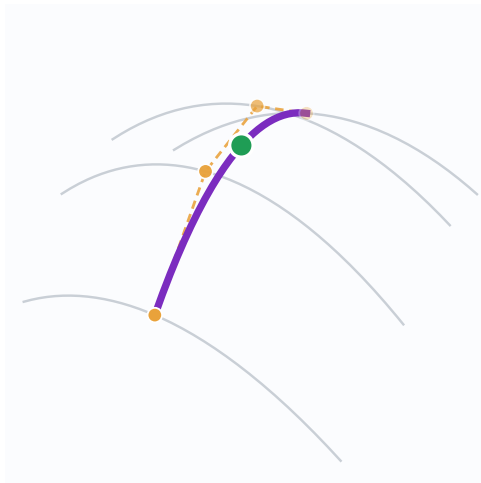
The de Casteljau Algorithm

These four points control the isoparametric curve in the v -direction.



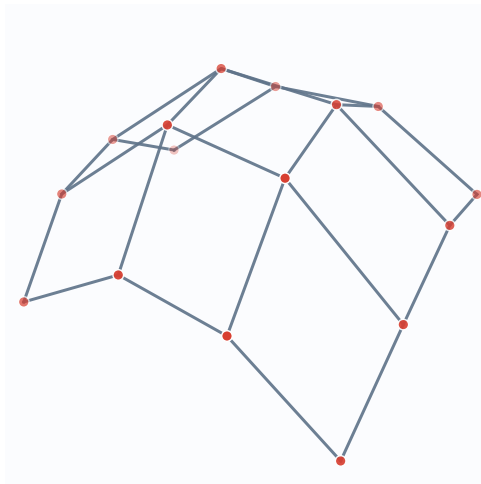
The de Casteljau Algorithm

Evaluating this curve at $v = v_0$ gives $\mathbf{P}(u_0, v_0)$.



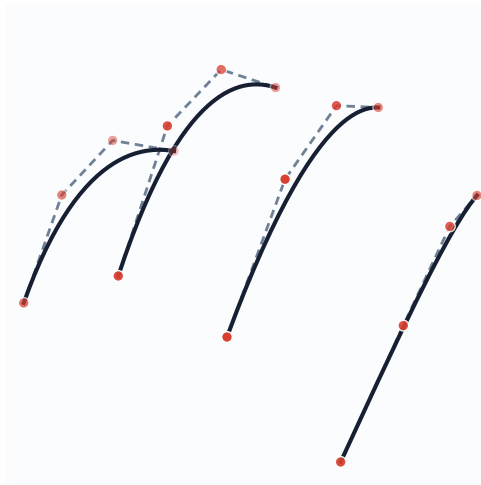
The de Casteljau Algorithm

The evaluation order may be reversed; restart from the same control net.



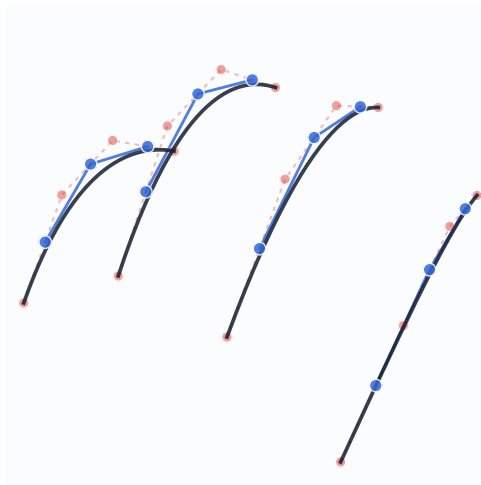
The de Casteljau Algorithm

Second order: expose the four control curves in the v -direction.



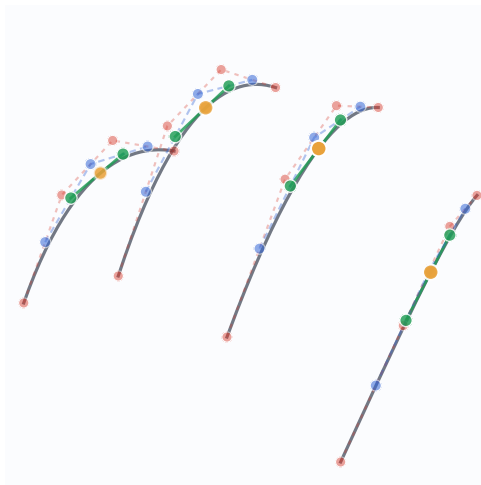
The de Casteljau Algorithm

Apply the first de Casteljau reduction at $v = v_0$ to all four curves.



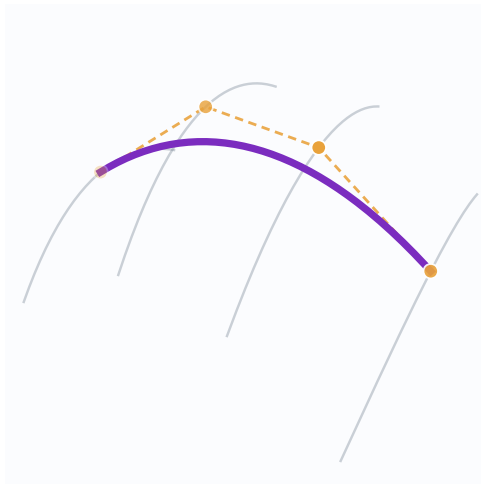
The de Casteljau Algorithm

Complete the four evaluations; the gold points form the transverse control polygon.



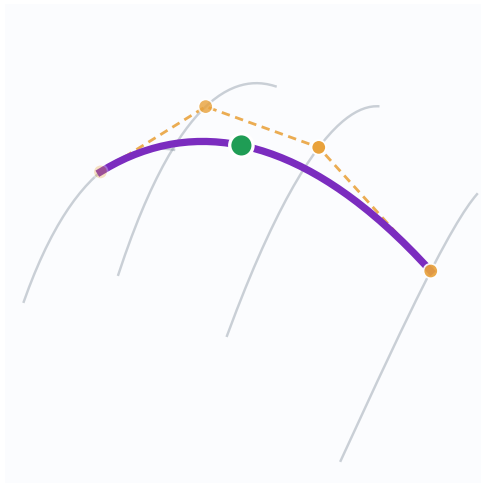
The de Casteljau Algorithm

These four points control the isoparametric curve in the u -direction.



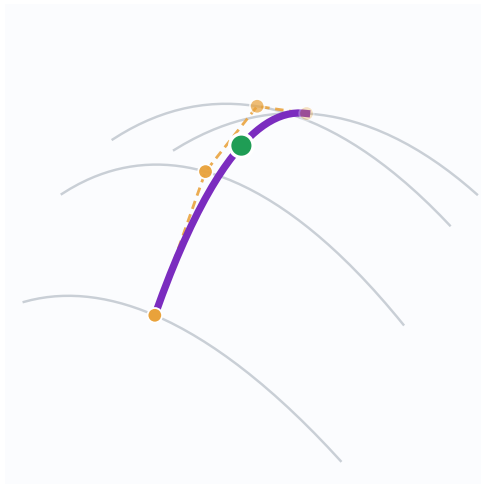
The de Casteljau Algorithm

Evaluating this curve at $u = u_0$ gives the same point $\mathbf{P}(u_0, v_0)$.

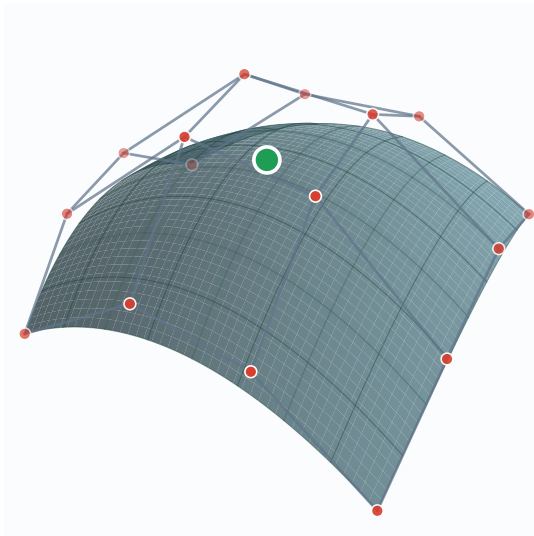


The de Casteljau Algorithm

Thus, the two evaluation orders agree.



A Bicubic Bézier Patch



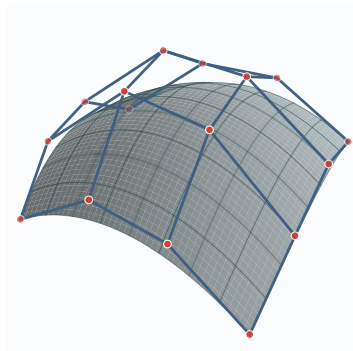
Properties of Tensor-Product Bézier Patches

- ▶ **Affine invariance** follows from the definition.
- ▶ **Convex hull property**: the patch lies in the convex hull of its control points.
- ▶ The patch interpolates the four corner control points:

$$\mathbf{P}(0, 0) = \mathbf{P}_{00}, \quad \mathbf{P}(1, 0) = \mathbf{P}_{m0},$$

$$\mathbf{P}(0, 1) = \mathbf{P}_{0n}, \quad \mathbf{P}(1, 1) = \mathbf{P}_{mn}.$$

- ▶ The variation-diminishing property is *not inherited* from the univariate case.



Degree Elevation

Goal

Rewrite a Bézier patch of degree (m, n) as one of degree $(m + 1, n)$.

Find the new coefficients such that

$$\mathbf{P}^{m,n}(u, v) = \sum_{j=0}^n \left[\sum_{i=0}^{m+1} \mathbf{P}_{ij}^{(1,0)} B_i^{m+1}(u) \right] B_j^n(v).$$

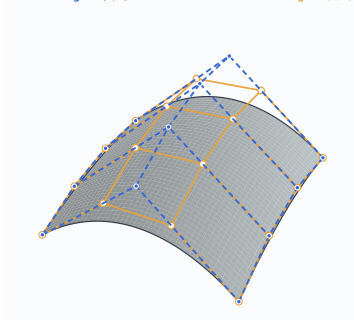
This reduces to a series of univariate problems.

$$\mathbf{P}_{ij}^{(1,0)} = \frac{i}{m+1} \mathbf{P}_{i-1,j} + \left(1 - \frac{i}{m+1}\right) \mathbf{P}_{ij}$$

where $i = 0, \dots, m+1, j = 0, \dots, n$ (terms with a zero coefficient are omitted).

3×4 net: degree (2, 3)

4×4 net: degree (3, 3)



Derivatives

As in the curve case, taking derivatives is accomplished by **differencing the control points**.

Partial derivative in the u -direction:

$$\frac{\partial \mathbf{P}^{m,n}}{\partial u}(u, v) = m \sum_{i=0}^{m-1} \sum_{j=0}^n \Delta^{1,0} \mathbf{P}_{ij} B_i^{m-1}(u) B_j^n(v), \quad \Delta^{1,0} \mathbf{P}_{ij} = \mathbf{P}_{i+1,j} - \mathbf{P}_{ij}.$$

More generally,

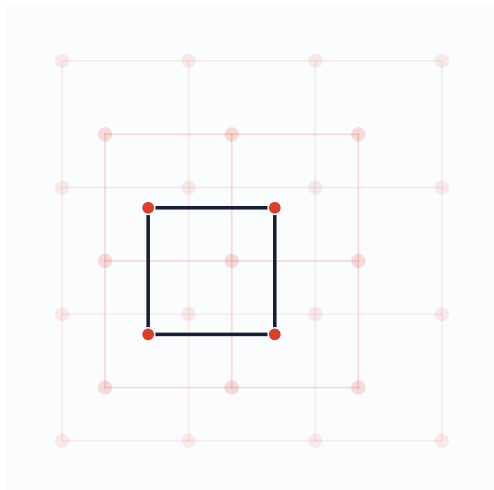
$$\frac{\partial^{r+s} \mathbf{P}^{m,n}}{\partial u^r \partial v^s}(u, v) = \frac{m!n!}{(m-r)!(n-s)!} \sum_{i=0}^{m-r} \sum_{j=0}^{n-s} \Delta^{r,s} \mathbf{P}_{ij} B_i^{m-r}(u) B_j^{n-s}(v).$$

Here $0 \leq r \leq m$, $0 \leq s \leq n$, and $\Delta^{0,0} \mathbf{P}_{ij} = \mathbf{P}_{ij}$.

$$\begin{aligned} \Delta^{r,s} \mathbf{P}_{ij} &= \Delta^{r-1,s} \mathbf{P}_{i+1,j} - \Delta^{r-1,s} \mathbf{P}_{ij} \quad (r \geq 1), \\ &= \Delta^{r,s-1} \mathbf{P}_{i,j+1} - \Delta^{r,s-1} \mathbf{P}_{ij} \quad (s \geq 1). \end{aligned}$$

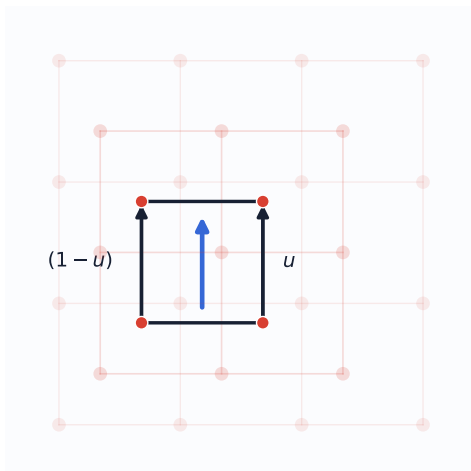
Derivatives Using de Casteljau's Algorithm

The final 2×2 net contains the two tangent directions.



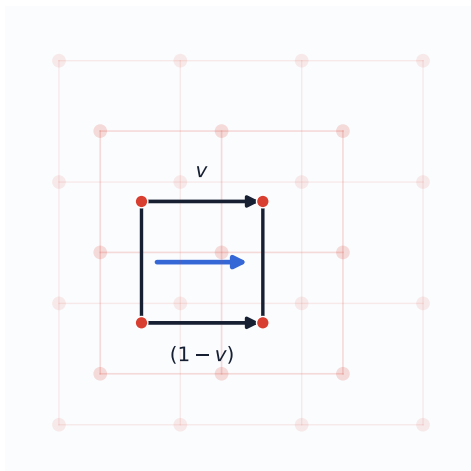
Derivatives Using de Casteljau's Algorithm

Vertical differences, combined with weights $(1 - u)$ and u , give the direction of $\partial \mathbf{P} / \partial v$ (up to the degree factor).



Derivatives Using de Casteljau's Algorithm

Horizontal differences, combined with weights $(1 - v)$ and v , give the direction of $\partial \mathbf{P} / \partial u$ (up to the degree factor).



Polar Forms of Tensor-Product Patches

$$\mathbf{P}(u_1, \dots, u_m \mid v_1, \dots, v_n)$$

- **Symmetry**: independent permutations of the u -arguments and v -arguments leave the polar values unchanged.
- **Multi-affinity**: for example, replacing u_k by $(1 - d)u_k + ds_k$ and v_ℓ by $(1 - e)v_\ell + ew_\ell$, i.e.

$$\mathbf{P}(\dots, (1 - d)u_k + ds_k, \dots \mid \dots, (1 - e)v_\ell + ew_\ell, \dots)$$

gives

$$\begin{aligned} & (1 - d)(1 - e) \mathbf{P}(\dots, u_k, \dots \mid \dots, v_\ell, \dots) \\ & + (1 - d)e \mathbf{P}(\dots, u_k, \dots \mid \dots, w_\ell, \dots) \\ & + d(1 - e) \mathbf{P}(\dots, s_k, \dots \mid \dots, v_\ell, \dots) \\ & + de \mathbf{P}(\dots, s_k, \dots \mid \dots, w_\ell, \dots). \end{aligned}$$

- **Diagonal**: $\mathbf{P}(u, \dots, u \mid v, \dots, v) = \mathbf{P}(u, v)$.

Example: Curves on Bézier Surfaces

Given two points in the domain of a Bézier patch of degree (n, n) ,

$$\mathbf{p} = (p_u, p_v), \quad \mathbf{q} = (q_u, q_v),$$

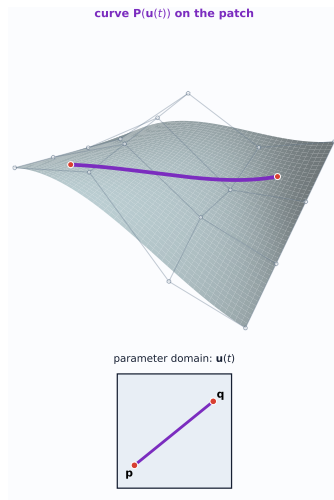
they define the straight line

$$\mathbf{u}(t) = (1 - t)\mathbf{p} + t\mathbf{q}.$$

This line is mapped to a curve on the surface. What are its Bézier control points?

Using the polar form

$$\mathbf{P}\left(\left((1-t)p_u + tq_u\right)^{\langle n \rangle} \mid \left((1-t)p_v + tq_v\right)^{\langle n \rangle}\right).$$



Example: Curves on Bézier Surfaces

Expanding both groups of u - and v -arguments gives a Bézier curve of degree at most $2n$:

$$\mathbf{P}(\mathbf{u}(t)) = \sum_{k=0}^{2n} \mathbf{c}_k B_k^{2n}(t).$$

Control points of the image curve

$$\mathbf{c}_k = \frac{1}{\binom{2n}{k}} \sum_{\substack{0 \leq i, j \leq n \\ i+j=k}} \binom{n}{i} \binom{n}{j} \mathbf{P}\left(p_u^{\langle n-i \rangle}, q_u^{\langle i \rangle} \mid p_v^{\langle n-j \rangle}, q_v^{\langle j \rangle}\right).$$

⇒ **More generally**, a degree- d curve in parameter space yields a curve of degree at most $(m+n)d$ on a patch of bidegree (m, n) ; equality holds generically.

Normal Vectors

The normal vector of a surface is the cross product of two tangent vectors.

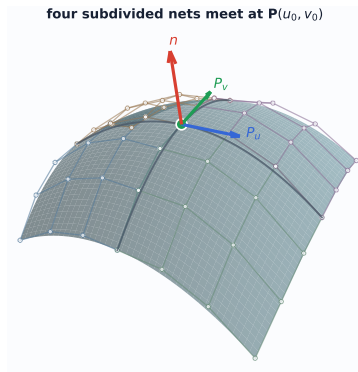
Using the partial derivatives

$$\mathbf{n}(u, v) = \frac{\frac{\partial \mathbf{P}}{\partial u}(u, v) \wedge \frac{\partial \mathbf{P}}{\partial v}(u, v)}{\left\| \frac{\partial \mathbf{P}}{\partial u}(u, v) \wedge \frac{\partial \mathbf{P}}{\partial v}(u, v) \right\|}.$$

At a corner, for instance,

$$\mathbf{n}(0, 0) = \frac{\Delta^{1,0} \mathbf{P}_{00} \wedge \Delta^{0,1} \mathbf{P}_{00}}{\left\| \Delta^{1,0} \mathbf{P}_{00} \wedge \Delta^{0,1} \mathbf{P}_{00} \right\|}.$$

Remark: Degenerate patches must be treated separately.



The Matrix Form of a Bézier Patch

The “geometry matrix” of the patch:

$$\mathbf{P}^{m,n}(u, v) = [B_0^m(u) \quad \cdots \quad B_m^m(u)] \begin{bmatrix} \mathbf{P}_{00} & \cdots & \mathbf{P}_{0n} \\ \vdots & & \vdots \\ \mathbf{P}_{m0} & \cdots & \mathbf{P}_{mn} \end{bmatrix} \begin{bmatrix} B_0^n(v) \\ \vdots \\ B_n^n(v) \end{bmatrix}.$$

Transformation to the monomial basis:

$$\mathbf{P}^{m,n}(u, v) = [u^0 \quad \cdots \quad u^m] M^T [\mathbf{P}_{ij}] N \begin{bmatrix} v^0 \\ \vdots \\ v^n \end{bmatrix},$$

where

$$M_{ij} = (-1)^{j-i} \binom{m}{j} \binom{j}{i} \quad (0 \leq i \leq j \leq m),$$

$$N_{ij} = (-1)^{j-i} \binom{n}{j} \binom{j}{i} \quad (0 \leq i \leq j \leq n).$$

Bézier Patch of a Graph

Consider a parameterization of the form

$$\mathbf{x}(u, v) = \begin{bmatrix} u \\ v \\ f(u, v) \end{bmatrix}.$$

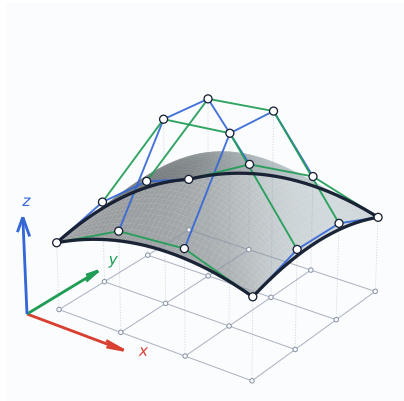
If

$$f(u, v) = \sum_{i=0}^m \sum_{j=0}^n p_{ij} B_i^m(u) B_j^n(v),$$

then the patch control points are

$$\mathbf{P}_{ij} = \begin{bmatrix} i/m \\ j/n \\ p_{ij} \end{bmatrix}.$$

They lie above a regular partition of the domain rectangle.



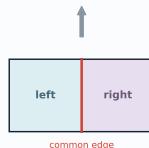
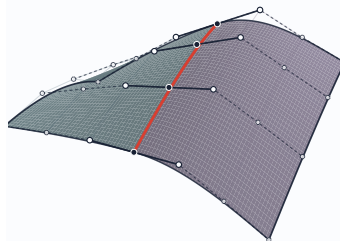
Composite Bézier Surfaces

- Suppose two Bézier patches of degree (m, n) are given, with
 left: $\{\mathbf{P}_{ij} : 0 \leq i \leq m, 0 \leq j \leq n\}$,
 right: $\{\mathbf{P}_{ij} : m \leq i \leq 2m, 0 \leq j \leq n\}$.
- To obtain C^r continuity across their common boundary, compare the u -partial derivatives up to order r , assuming the same boundary parameterization and scale.

Equivalent curve condition

Each row crossing the join defines two Bézier curve segments that must meet with C^r continuity.

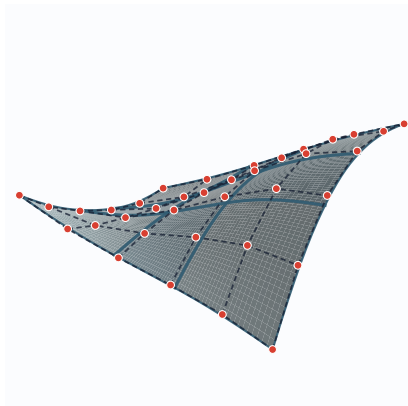
adjacent control polygons across the join



Tensor-Product B-spline Surfaces

$$\mathbf{P}(u, v) = \sum_i \sum_j \mathbf{P}_{ij} N_i^m(u) N_j^n(v).$$

- ▶ Two knot sequences are needed: one in the u -direction and one in the v -direction.
- ▶ All methods and algorithms reduce to the curve case, applied successively in each direction.

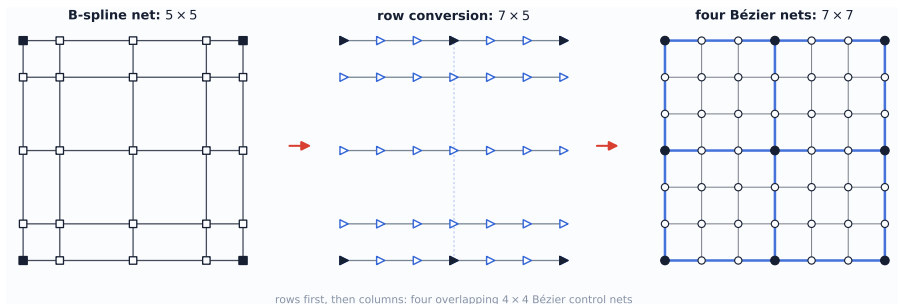


A bicubic B-spline surface consisting of 3×3 Bézier patches.

Conversion to Bézier Patches

Conversion is performed using the univariate method:

- ▶ interpret the B-spline control net row by row as univariate B-spline polygons and convert them to piecewise Bézier form;
- ▶ interpret the resulting Bézier points column by column as B-spline polygons, then convert each of them to Bézier form.



Rational Bézier and B-spline Surfaces

- ▶ Bézier and B-spline surfaces generalize to rational surfaces exactly as curves do.
- ▶ They are projections of four-dimensional tensor-product surfaces.
- ▶ Standard NURBS use positive weights; more generally, the denominator must not vanish on the parameter domain.

Rational Bézier surface

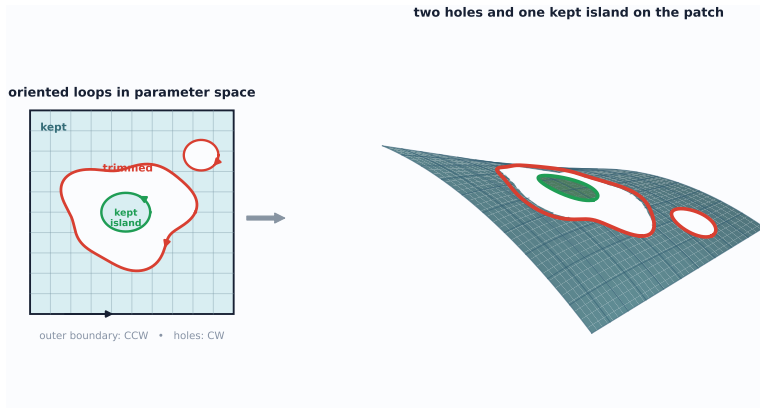
$$\mathbf{P}(u, v) = \frac{\sum_i \sum_j w_{ij} \mathbf{P}_{ij} B_i^m(u) B_j^n(v)}{\sum_i \sum_j w_{ij} B_i^m(u) B_j^n(v)}.$$

Rational B-spline surface

$$\mathbf{P}(u, v) = \frac{\sum_i \sum_j w_{ij} \mathbf{P}_{ij} N_i^m(u) N_j^n(v)}{\sum_i \sum_j w_{ij} N_i^m(u) N_j^n(v)}.$$

Trimmed Surfaces

- ▶ A parametric curve in the domain trims the surface; a degree- d curve yields a curve of degree at most $(m+n)d$ on a patch of bidegree (m, n) .
- ▶ Curve orientation defines inside and outside, including for nested loops; classification can be performed by a ray-crossing test.



B-rep of CAD Models

Boundary representation

A CAD model is assembled from trimmed parametric faces; the underlying tensor-product surfaces generally extend beyond the visible boundaries.

